Homework 13

Zhiqi Gan

Problem 13.1.
Solution 13.1.
1. $\operatorname{trace}(A)=11, \det(A)=28 \implies b=11, a=28.$

2. By the big formula, $\det(A)=a=24, \operatorname{trace}(A)=c=9$. Now $A=\begin{bmatrix}0&0&24\\1&0&b\\0&1&9\end{bmatrix}$. We also have $\det(A-3I)=0 \implies \begin{vmatrix}-3&0&24\\1&-3&b\\0&1&6\end{vmatrix}=0 \implies b=-26.$

3. $\det(A)=-16 \implies \det(B)=\det(A)=-16 \implies y=-4 \implies \operatorname{trace}(B)=0 \implies \operatorname{trace}(A)=0 \implies x=-2.$

4. $\det(A)\neq 0 \implies a\neq \frac{2}{3}$. We also have $A\boldsymbol{x}=\lambda\boldsymbol{x}$, then $\begin{cases}2+b+1=\lambda\\1+2b+1=\lambda b\\1+b+a=\lambda\end{cases}$. The solution is $b=1, a=2$.

5. By observation, $\det(A)=1, \operatorname{trace}(A)=-1$, then the eigenvalues are $\lambda_1=\lambda_2=-1, \lambda_3=1$. Alternatively, you can also solve the characteristic polynomial. Since $\lambda=1$ is simple, it can not be defective. We only need to check $\lambda=-1$, whose algebraic multiplicity is 2. So A is diagonalizable if and only if the geometric multiplicity of -1 is 2.

$$M_g(-1)=\dim\ker(A-(-1)I)=3-\operatorname{rank}(A+I)=2 \implies \operatorname{rank}(A+I)=1$$

But $\begin{bmatrix}3&2&-2\\-k&-1&k\\4&2&-3\end{bmatrix}+I=\begin{bmatrix}4&2&-2\\-k&0&k\\4&2&-2\end{bmatrix}$. The rank of the matrix is 1 if and only if $k=0$.

Problem 13.2.
Solution 13.2.
1. eigenvalues $0,0,1,2$.
2. eigenvalues $1,1,4,9$.
3. eigenvalues $1,1,\frac{1}{2},\frac{1}{3}$.

Lemma 1.
Set the eigenvalues of A are λ, then the eigenvalues of $\overline{A}$ are $\overline{\lambda}$.
Proof:
Set the eigenvalues of A are λ, the corresponding eigenvectors are $\boldsymbol{v}$, then

$$A\boldsymbol{v}=\lambda\boldsymbol{v} \implies \overline{A\boldsymbol{v}}=\overline{\lambda\boldsymbol{v}} \implies \overline{A}\overline{\boldsymbol{v}}=\overline{\lambda}\overline{\boldsymbol{v}}$$

Hence the eigenvalues of $\overline{A}$ are $\overline{\lambda}$.
Lemma 1 proof ends.

4. According to Lemma 1, the eigenvalues of $\overline{A}$ are $1,1,2,3$.

Lemma 2.
Set the eigenvalues of A are λ, then the eigenvalues of A^* are $\overline{\lambda}$.
Proof:
λ is an eigenvalue if and only if $\det(A-\lambda I)=0$. So

$$\det\big((A-\lambda I)^*\big)=\det\Big(\overline{(A-\lambda I)}^{\mathrm{T}}\Big)=\det\Big(\overline{A-\lambda I}\Big)$$

Since determinant is a polynomial about all the entries of $A-\lambda I$,

$$\det\Big(\overline{A-\lambda I}\Big)=\overline{\det(A-\lambda I)} \implies \det\big((A-\lambda I)^*\big)=\overline{\det(A-\lambda I)}$$

Therefore, λ is an eigenvalue of $A \iff \det(A-\lambda I)=0 \iff \det\big((A-\lambda I)^*\big)=\overline{0}=0 \iff \det\big(A^*-(\lambda I)^*\big)=0 \iff \det\Big(A^*-\overline{\lambda}I\Big)=0 \iff \overline{\lambda}$ is an eigenvalue of A^*.
Lemma 2 proof ends.

5. According to Lemma 2, the eigenvalues of A^* are $1,1,2,3$.
6. eigenvalues $0,0,0,0$.

Lemma 3.
The eigenvalues of $\begin{bmatrix} A & B \\ & C \end{bmatrix}$ are the combined eigenvalues of A and C.
Proof:

$$\det\left(\begin{bmatrix} A & B \\ & C \end{bmatrix} - \lambda I_{m+n}\right) = \det\left(\begin{bmatrix} A - \lambda I_m & B \\ & C - \lambda I_n \end{bmatrix}\right) = \det(A - \lambda I_m)\det(C - \lambda I_n)$$

So $\det\left(\begin{bmatrix} A & B \\ & C \end{bmatrix} - \lambda I_{m+n}\right) = 0 \iff \det(A - \lambda I_m) = 0$ or $\det(C - \lambda I_n) = 0 \iff \lambda$ is an eigenvalue of A or λ is an eigenvalues of C.
Lemma 3 proof ends.

7. According to Lemma 3, the eigenvalues of $\begin{bmatrix} A & \\ & A \end{bmatrix}$ are $1, 1, 2, 3, 1, 1, 2, 3$.
8. According to Lemma 3, the eigenvalues of $\begin{bmatrix} A + I & A \\ & A - I \end{bmatrix}$ are $2, 2, 3, 4, 0, 0, 1, 2$.

Problem 13.3.
Solution 13.3.
1.

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 5 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}^{1024} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{1024} & \\ & 5^{1024} \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{2}{3}2^{1024} + \frac{1}{3}5^{1024} & \frac{2}{3}(5^{1024} - 2^{1024}) \\ \frac{1}{3}(5^{1024} - 2^{1024}) & \frac{1}{3}2^{1024} + \frac{2}{3}5^{1024} \end{bmatrix}$$

All entries $\geq \frac{1}{3}(5^{1024} - 2^{1024}) > 10^{700}$.
2.

$$\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix} = X \begin{bmatrix} i & \\ & -i \end{bmatrix} X^{-1}$$

$$\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix}^{1024} = X \begin{bmatrix} i^{1024} & \\ & (-i)^{1024} \end{bmatrix} X^{-1} = X \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} X^{-1} = I_2$$

3.

$$\begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix} = X \begin{bmatrix} \frac{-1+\sqrt{3}i}{2} & \\ & \frac{-1-\sqrt{3}i}{2} \end{bmatrix} X^{-1}$$

Note that $(\frac{-1+\sqrt{3}i}{2})^3 = (\frac{-1-\sqrt{3}i}{2})^3 = 1$, then

$$\begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}^{1024} = X \begin{bmatrix} (\frac{-1+\sqrt{3}i}{2})^{1024} & \\ & (\frac{-1-\sqrt{3}i}{2})^{1024} \end{bmatrix} X^{-1} = X \begin{bmatrix} \frac{-1+\sqrt{3}i}{2} & \\ & \frac{-1-\sqrt{3}i}{2} \end{bmatrix} X^{-1} = \begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}$$

Problem 13.4.
Solution 13.4.
1. eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 10$. $M_a(10) = 3, M_g(10) = 1$. Not diagonalizable.
2. eigenvalues $\lambda_1 = 10, \lambda_2 = 10.001, \lambda_3 = 10.002$. All eigenvalues are simple. $M_a = M_g = 1$. Diagonalizable.
3. eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$. $M_a(0) = 4, M_g(0) = 1$. Not diagonalizable.
4. $\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 & & \\ & \lambda & -1 & \\ & & \lambda & -1 \\ -10^{-4} & & & \lambda \end{vmatrix} = \lambda^4 - 10^{-4}$. So eigenvalues are $\lambda_1 = 10^{-1}, \lambda_2 = -1 \times 10^{-1}, \lambda_3 = i \times 10^{-1}, \lambda_4 = -i \times 10^{-1}$. All eigenvalues are simple. $M_a = M_g = 1$. Diagonalizable.

Problem 13.5.
Solution 13.5.
1.
Method 1:
By observation, one of eigenvalues is $a - b$.

$$M_g(a - b) = \dim\ker(A - (a - b)I) = n - \operatorname{rank}(A - (a - b)I) = n - 1$$

Therefore, $M_a(a-b) \geq n-1$. However, $\operatorname{trace}(A) = na$, so we have another simple eigenvalue $na-(n-1)(a-b) = a+(n-1)b$. Thus, $M_g(a-b) = M_a(a-b) = n-1, M_g(a+(n-1)b) = M_a(a+(n-1)b) = 1$. The matrix is diagonalizable.

$$A = \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} a-b & & & \\ & \ddots & & \\ & & a-b & \\ & & & a+(n-1)b \end{bmatrix} \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix}^{-1}$$

Method 2:

Set $X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$, then $A = bX + (a-b)I$. Note that X is diagonalizable and its eigenvalues are $\lambda_1 = \cdots = \lambda_{n-1} = 0, \lambda_n = n$. So the diagonalization of X is

$$X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & 0 & \\ & & & n \end{bmatrix} \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix}^{-1}$$

Since $A = bX + (a-b)I$, and it's a polynomial about X, the eigenvalues of A are $\lambda_1 = \cdots = \lambda_{n-1} = a-b, \lambda_n = nb+a-b$. So the diagonalization of A is

$$A = \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} a-b & & & \\ & \ddots & & \\ & & a-b & \\ & & & nb+a-b \end{bmatrix} \begin{bmatrix} 1 & & & 1 \\ -1 & 1 & & 1 \\ & \ddots & \ddots & \vdots \\ & & -1 & 1 \end{bmatrix}^{-1}$$

2. $\lambda_1 = \lambda_2 = -1, M_a(-1) = 2, M_g(-1) = 1$. Not diagonalizable.
Schur Decomposition:

$$A = \begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ -1 & 1 \end{bmatrix}$$

Problem 13.6.
Solution 13.6.
1. Obviously, $\lambda_1 = 0, \lambda_2 = \frac{15+3\sqrt{33}}{2}, \lambda_3 = \frac{15-3\sqrt{33}}{2}$. We have three distinct eigenvalues, so the matrix is diagonalizable.
2. $\lambda_1 = \lambda_2 = 0, \lambda_3 = 1$. So $M_a(0) = 2$. At the same time $M_g(0) = \dim\ker(A - \lambda I) = \dim\ker(A) = n - \operatorname{rank}(A) = 2$. So the matrix is diagonalizable.
3. $\operatorname{rank}(A) = 1$, so $\ker(A) = n-1 \implies$ geometric multiplicity $M_g(0) = n-1 \implies$ algebraic multiplicity $M_a(0) \geq n-1$. So we have at least $n-1$ eigenvalues are 0. The remaining one we don't know, let's say it's λ_n. According to Homework 6.1.6, $\operatorname{trace}(A) = \operatorname{trace}(\boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}) = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = 0$. Hence $\operatorname{trace}(A) = 0 + \cdots + 0 + \lambda_n = 0 \implies \lambda_n = 0 \implies M_a(0) = n \implies M_g(0) \neq M_a(0)$. So it's not diagonalizable.
4. Similar to subproblem 3, the matrix is diagonalizable.

Problem 13.7.
Solution 13.7.
1.
Method 1:
$A^2 - 4I = O \implies (A+2I)(A-2I) = O \implies \operatorname{Ran}(A-2I) \subseteq \ker(A+2I) \implies \operatorname{rank}(A-2I) \leq \dim\ker(A+2I) \implies n - \dim\ker(A-2I) \leq \dim\ker(A+2I)$. Hence we have

$$\dim\ker(A+2I) + \dim\ker(A-2I) \geq n \tag{1}$$

Set $\boldsymbol{x} \in \ker(A+2I) \cap \ker(A-2I)$, then $(A+2I)\boldsymbol{x} = (A-2I)\boldsymbol{x} = \boldsymbol{0} \iff \boldsymbol{x} = \boldsymbol{0}$. So

$$\ker(A+2I) \cap \ker(A-2I) = \{\boldsymbol{0}\} \tag{2}$$

Combine equation (1) and (2), we have $\ker(A+2I) + \ker(A-2I) = \mathbb{C}^n$.
Set the eigenvectors of A is $\boldsymbol{v}$, $A^2 - 4I = O \implies A^2\boldsymbol{v} - 4\boldsymbol{v} = \boldsymbol{0} \implies \lambda^2\boldsymbol{v} - 4\boldsymbol{v} = \boldsymbol{0} \implies \lambda^2 - 4 = 0 \implies \lambda = \pm 2$.

Basis of $\ker(A+2I)$ and $\ker(A-2I)$ are basis of eigenspaces. So we have n linearly independent eigenvectors. A is diagonalizable.
Method 2:

$$A^2-4I=O \implies (A+2I)(A-2I)=(A-2I)(A+2I)=O \implies \mathrm{Ran}(A-2I)\subseteq\ker(A+2I), \mathrm{Ran}(A+2I)\subseteq\ker(A-2I)$$

By observation, $\forall \boldsymbol{v}\in\mathbb{C}^n$,

$$\boldsymbol{v}=\frac{1}{4}(A+2I)\boldsymbol{v}-\frac{1}{4}(A-2I)\boldsymbol{v}$$

And

$$\begin{aligned}\frac{1}{4}(A+2I)\boldsymbol{v}\in\mathrm{Ran}(A+2I) &\implies \frac{1}{4}(A+2I)\boldsymbol{v}\in\ker(A-2I)\\ -\frac{1}{4}(A-2I)\boldsymbol{v}\in\mathrm{Ran}(A-2I) &\implies \frac{1}{4}(A-2I)\boldsymbol{v}\in\ker(A+2I)\end{aligned}$$

Thus, any vectors in $\mathbb{C}^n$ can be represented as the sum of two vectors from $\ker(A-2I)$ and $\ker(A+2I)$, respectively. So $\ker(A+2I)+\ker(A-2I)=\mathbb{C}^n$.
And similar to method 1, we have n linearly independent eigenvectors. A is diagonalizable.
2.
Method 1:
$A^2-3A+2I=O \implies (A-I)(A-2I)=O \implies \mathrm{Ran}(A-2I)\subseteq\ker(A-I) \implies \mathrm{rank}(A-2I)\le\dim\ker(A-I) \implies n-\dim\ker(A-2I)\le\dim\ker(A-I)$. Hence we have

$$\dim\ker(A-I)+\dim\ker(A-2I)\ge n \tag{3}$$

Similar to subproblem 1, we also have

$$\ker(A-I)\cap\ker(A-2I)=\{\boldsymbol{0}\} \tag{4}$$

Combine equation (3) and (4), we have $\ker(A-I)+\ker(A-2I)=\mathbb{C}^n$.
Also similar to subproblem 1, the eigenvalues of A are 1 and 2, So we have n linearly independent eigenvectors. A is diagonalizable.
Method 2:

$$A^2-3A+2I=O \implies (A-I)(A-2I)=(A-2I)(A-I)=O \implies \mathrm{Ran}(A-2I)\subseteq\ker(A-I), \mathrm{Ran}(A-I)\subseteq\ker(A-2I)$$

By observation, $\forall \boldsymbol{v}\in\mathbb{C}^n$,

$$\boldsymbol{v}=(A-I)\boldsymbol{v}-(A-2I)\boldsymbol{v}$$

And

$$\begin{aligned}(A-I)\boldsymbol{v}\in\mathrm{Ran}(A-I) &\implies (A-I)\boldsymbol{v}\in\ker(A-2I)\\ -(A-2I)\boldsymbol{v}\in\mathrm{Ran}(A-2I) &\implies -(A-2I)\boldsymbol{v}\in\ker(A-I)\end{aligned}$$

Thus, any vectors in $\mathbb{C}^n$ can be represented as the sum of two vectors from $\ker(A-2I)$ and $\ker(A-I)$, respectively. So $\ker(A-I)+\ker(A-2I)=\mathbb{C}^n$.
And similar to method 1, we have n linearly independent eigenvectors. A is diagonalizable.
3.

$$\begin{aligned}\boldsymbol{v}&=aA(A-I)\boldsymbol{v}+bA(A+I)\boldsymbol{v}+c(A-I)(A+I)\boldsymbol{v}\\&=(aA^2-aA+bA^2+bA+cA^2-cI)\boldsymbol{v}\\&=((a+b+c)A^2+(b-a)A-cI)\boldsymbol{v}\end{aligned}$$

Therefore, $\begin{cases}a+b+c=0\\ b-a=0\\ -c=1\end{cases} \implies \begin{cases}a=\frac{1}{2}\\ b=\frac{1}{2}\\ c=-1\end{cases}$.
4.

$$A^3=A \implies A^3-A=O \implies A(A-I)(A+I)=(A+I)A(A-I)=(A-I)A(A+I)=O$$

Therefore,

$$\mathrm{Ran}((A-I)(A+I))\subseteq\ker(A) \tag{5}$$
$$\mathrm{Ran}(A(A-I))\subseteq\ker(A+I) \tag{6}$$
$$\mathrm{Ran}(A(A+I))\subseteq\ker(A-I) \tag{7}$$

From subproblem 3, $\forall \boldsymbol{v} \in \mathbb{C}^n$, we always have

$$\boldsymbol{v} = \frac{1}{2}A(A-I)\boldsymbol{v} + \frac{1}{2}A(A+I)\boldsymbol{v} - (A-I)(A+I)\boldsymbol{v}$$

where

$$\frac{1}{2}A(A-I)\boldsymbol{v} \in \mathrm{Ran}(A(A-I)) \tag{8}$$

$$\frac{1}{2}A(A+I)\boldsymbol{v} \in \mathrm{Ran}(A(A+I)) \tag{9}$$

$$-(A-I)(A+I)\boldsymbol{v} \in \mathrm{Ran}((A-I)(A+I)) \tag{10}$$

Combine equations (5)(6)(7) and (8)(9)(10),

$$\frac{1}{2}A(A-I)\boldsymbol{v} \in \ker(A+I)$$

$$\frac{1}{2}A(A+I)\boldsymbol{v} \in \ker(A-I)$$

$$-(A-I)(A+I)\boldsymbol{v} \in \ker(A)$$

Thus, any vectors in $\mathbb{C}^n$ can be represented as the sum of three vectors from $\ker(A+I), \ker(A-I), \ker(A)$, respectively. So $\ker(A+I) + \ker(A-I) + \ker(A) = \mathbb{C}^n$.